# Class 15

**Theorem** For every finite game, there is an NE in mixed strategies.

**Proof (Contd.)** $G = \langle N, (S_i)_{i \in N}, (U_i)_{i \in N} \rangle$

$|N| = n$

$|S_i| = m_i \quad \forall i = 1(1)n$

$g_i^j : \Sigma \to \mathbb{R} \quad \forall i = 1(1)n$

$\forall j = 1(1)m_i$

$$g_i^j(\sigma) = \max\{0, U_i(s_i^j, \sigma_{-i}) - U_i(\sigma)\}$$

**Claim** $g_i^j$ is a Cont function of $\sigma$ $\forall i = 1(1)n$ and $\forall j = 1(1)m_i$.

**Claim:** $\sigma$ is an NE iff $g_i^j(\sigma) = 0$ $\forall i = 1(1)n$ $\forall j = 1(1)m_i$

**Our goal** $f : \Sigma \to \Sigma$

- $f$ is Cont
- every fixed point of $f$ is an NE of the game.

**Construction of $f$**

$$f_i^j(\sigma) = \sigma_i(s_i^j) + g_i^j(\sigma)$$

$$\frac{}{1+\sum_{j=1}^{m_i} g_i^j(\sigma)} \qquad \forall i=1(1)n \quad \forall j=1(1)m_i$$

**Claim** $f(\sigma) \in \Sigma$

- $f_i^j(\sigma) \geq 0 \quad \forall i=1(1)n \quad \forall j=1(1)m_i \quad \forall \sigma \in \Sigma.$
- $\sum_{j=1}^{m_i} f_i^j(\sigma) = 1 \quad \forall i=1(1)n \quad \forall \sigma \in \Sigma.$

**Proof** Verify.

**Claim** $f$ is cont function of $\sigma$.

**Proof** as $g_i^j$ s are cont functions of $\sigma$.

**Claim** Let $\sigma$ be a fixed point of $f$. Then

$$g_i^j(\sigma) = \sigma_i(s_i^j) \sum_{k=1}^{m_i} g_i^k(\sigma) \qquad \forall i=1(1)n \quad \forall j=1(1)m_i$$

$f((\sigma_1, \sigma_2)) = (\sigma_1', \sigma_2')$

**Proof** Suppose $\sigma$ is a fixed point

$$f_i^j(\sigma) = \sigma_i(s_i^j) \qquad \forall i=1(1)n \quad \forall j=1(1)m_i$$

$$\Rightarrow \frac{\sigma_i(s_i^j) + g_i^j(\sigma)}{1+\sum_{k=1}^{m_i} g_i^k(\sigma)} = \sigma_i(s_i^j)$$

$$\Rightarrow g_i^j(\sigma) = \sigma_i(s_i^j) \sum_{k=1}^{m_i} g_i^k(\sigma).$$

**Claim** If $\sigma$ is a fixed point then $\sigma$ is an NE of $G$.

**Proof** Assume for contradiction $\sigma$ is not an NE.

$\exists p \in N$ and $a_p \in \{1, \dots, m_p\}$ s.t.

$$g_p^a(\sigma) > 0 \implies \sum_{k=1}^{m_p} g_p^k(\sigma) > 0$$

As $\sigma$ is a fixed point

$$g_i^j(\sigma) = \sigma_i(s_i^j) \sum_{k=1}^{m_i} g_i^k(\sigma) \qquad \forall i = 1(1)n, \quad \forall j = 1(1)m_i$$

$$\implies g_p^j(\sigma) = \sigma_p(s_p^j) \sum_{k=1}^{m_p} g_p^k(\sigma)$$

$g_p^j(\sigma) > 0 \iff \sigma_p(s_p^j) > 0. \quad \forall j = 1(1)m_p$

$$U_p(\sigma) = \sum_{k=1}^{m_p} \sigma_p(s_p^k)\, U(s_p^k, \sigma_{-p})$$

$$\implies \sum_{k=1}^{m_p} \sigma_p(s_p^k) \left( U(s_p^k, \sigma_{-p}) - U_p(\sigma) \right) = 0$$

$$\implies \sum_{\{k \mid \sigma_p(s_p^k) > 0\}} \sigma_p(s_p^k) \left( U(s_p^k, \sigma_{-p}) - U_p(\sigma) \right) = 0$$

$$\implies \sum_{\{k \mid \sigma_p(s_p^k) > 0\}} \sigma_p(s_p^k)\, g_p^k(\sigma) = 0$$

$g_p^k(\sigma) = \max\{0, (U(s_p^k, \sigma_{-p}) - U_p(\sigma))\}$

As $\sigma_p(s_p^k) > 0 \iff g_p^k(\sigma) > 0$

$$\sum_{\{k \mid \sigma_p(s_p^k) > 0\}} \sigma_p(s_p^k)\, g_p^k(\sigma) > 0$$ , so we have a contradiction. Thus means $\sigma$ is an NE.

$\sigma^*$ is an NE

$U_i(\sigma^*)$

$\geq U_i(\sigma_i, \sigma_{-i}^*)$

$\forall \sigma_i \in \Sigma_i$

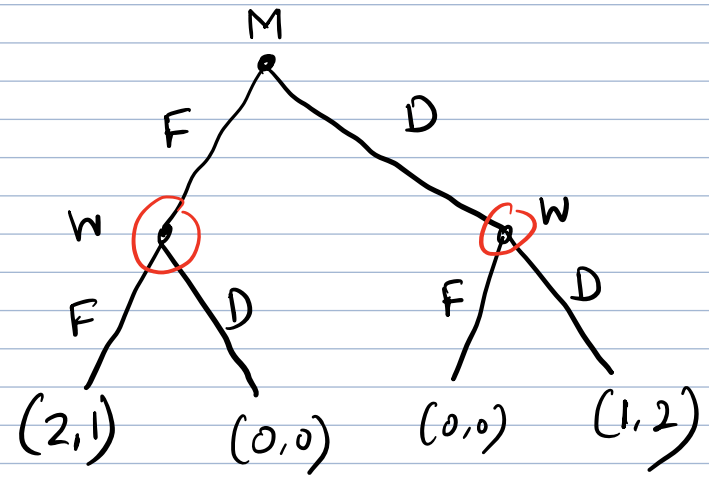
$\Leftrightarrow U_i(\sigma^*)$

$\geq U_i(s_i, \sigma_{-i}^*)$

$\forall s_i \in S_i$

## Extensive form games

- The man takes his action first and the the human sees the choice and make hers own move.




M → F , W → F

M → D , W → D

( M → F , W → F )

| | FF | FD | DF | DD |
|---|---|---|---|---|
| F | (2,1) | (2,1) | (0,0) | (0,0) |
| D | (0,0) | (1,2) | (0,0) | (1,2) |